

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA15

COURSE OUTCOME COVERED

CO4: Analyze big data characteristics and challenges across domains including security and application aspects. (BL4)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Articulation (Level 4)
Question Count: 5

Total Marks: 40

Code of Conduct

I **K.MAHESH BABU(192525122)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **K.MAHESH BABU(192525122)**

Assessment Tasks

Industry Problem Solving Task

1. An e-commerce company receives clickstream, transaction, and review data from millions of users daily. Propose a big data analytics pipeline and explain how the 5Vs influence architecture design.
2. A healthcare organization collects structured patient records, semi-structured insurance claims, and unstructured medical images. Design an approach to manage and secure this data ecosystem.
3. A telecom company wants to detect churn patterns using high-volume customer interaction data. Identify the big data challenges and recommend suitable analytics and integration strategies.
4. A smart city project generates sensor streams from traffic, air quality, and surveillance systems. Explain how distributed file systems and data integration tools can support large-scale analysis.
5. A financial analytics firm must comply with strict privacy requirements while processing high-velocity transaction data. Design a secure big data solution and justify the controls chosen.

Industry Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application

Solution Design	25%	Solution is practical, scalable, and well reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

1. E-Commerce Company – Big Data Analytics Pipeline

Question

An e-commerce company receives clickstream, transaction, and review data from millions of users daily. Propose a Big Data analytics pipeline and explain how the 5Vs influence architecture design.

Answer

An e-commerce company generates a huge amount of data from customer activities such as website clicks, product searches, purchases, payment transactions, ratings, and reviews. A scalable Big Data analytics pipeline is required to collect, store, process, analyze, and visualize this information.

Proposed Big Data Pipeline

Users → Data Sources → Data Ingestion → Distributed Storage → Data Processing → Analytics →

Visualization/Decision Making

1. Data Sources

The main sources of data are:

- Clickstream data from websites and mobile applications
- Customer transaction records
- Product searches
- Customer reviews and ratings
- Payment information
- Customer profiles
- Product information

2. Data Ingestion

Data ingestion tools collect data continuously from different sources. Real-time clickstream and transaction data can be sent to a streaming system, while older records can be transferred through batch processing.

3. Distributed Storage

The collected data should be stored in a scalable distributed storage system. Structured transaction data can be stored in databases, while large logs, reviews, and other datasets can be stored in distributed file or object storage.

4. Data Processing

The stored data can be processed using distributed processing frameworks. Processing can include:

- Removing duplicate records
- Cleaning incorrect data
- Combining information from different sources
- Calculating customer activity
- Identifying popular products
- Analyzing purchasing patterns

5. Analytics

Analytics can be used for:

- Product recommendations
- Customer behavior analysis
- Sales forecasting
- Fraud detection
- Customer segmentation
- Personalized advertisements

6. Visualization

Dashboards and reports can display sales trends, customer activity, product performance, and business indicators to managers.

Influence of the 5Vs

5V	Influence on Architecture
Volume	Millions of users generate huge datasets, so scalable distributed storage is required.

Velocity	Clicks and transactions arrive continuously, requiring real-time or near-real-time processing.
Variety	Transactions, reviews, images, logs, and clickstream data require support for different formats.
Veracity	Incorrect reviews, duplicate transactions, or incomplete customer information must be cleaned and validated.
Value	The architecture should convert raw data into useful information such as recommendations, sales insights, and customer trends.

A distributed and scalable pipeline allows the e-commerce company to handle millions of users and continuously growing data. The 5Vs directly influence storage capacity, processing speed, data formats, data quality mechanisms, and analytical requirements.

2. Healthcare Organization – Data Management and Security

Question

A healthcare organization collects structured patient records, semi-structured insurance claims, and unstructured medical images. Design an approach to manage and secure this data ecosystem.

Answer

A healthcare organization handles different types of sensitive information. Therefore, the system must support different data formats while providing strong security, privacy, reliability, and controlled access.

Types of Data

1. Structured Data

Patient records stored in tables are structured data.

Examples:

- Patient ID
- Name
- Age
- Blood group
- Diagnosis
- Admission date
- Prescription details

This data can be stored in relational databases.

2. Semi-Structured Data

Insurance claims may contain flexible fields and metadata.

Examples:

- JSON claims
- XML documents
- Electronic forms

These can be stored using suitable document-oriented or distributed storage systems.

3. Unstructured Data

Medical images such as X-rays, CT scans, and MRI scans are unstructured data.

These require high-capacity storage and specialized processing systems.

Proposed Architecture

Patient Systems → Data Integration → Secure Data Ingestion → Distributed Storage → Processing/Analytics → Authorized Users

Data Integration

Integration tools can collect information from:

- Hospital databases
- Insurance systems
- Medical imaging systems
- Laboratory systems
- Patient monitoring systems

The data should be cleaned and transformed before analysis.

Security Measures

1. Authentication

Only authorized users should be allowed to access the system.

2. Authorization

Different users should have different permissions. For example, a doctor may access patient medical records, while an administrative employee may only access billing information.

3. Encryption

Sensitive data should be encrypted during storage and transmission.

4. Access Control

Role-based access control can restrict users according to their responsibilities.

5. Audit Logs

All important access and modification activities should be recorded.

6. Backup and Recovery

Regular backups should be maintained so that data can be recovered after accidental deletion, hardware failure, or other incidents.

Data Management

A proper data-management strategy should include:

- Data classification
- Data cleaning
- Metadata management
- Data quality checking
- Controlled access
- Backup
- Retention policies

Conclusion

The healthcare organization should use a combination of relational databases, flexible data storage, and high-capacity storage for different data types. Strong authentication, authorization, encryption, auditing, and backup mechanisms should protect the sensitive healthcare ecosystem.

3. Telecom Company – Customer Churn Detection

Question

A telecom company wants to detect churn patterns using high-volume customer interaction data. Identify the Big Data challenges and recommend suitable analytics and integration strategies.

Answer

Customer churn occurs when customers stop using a telecom company's services. A telecom company generates large amounts of customer interaction data from calls, messages, internet usage, complaints, recharge history, service requests, and billing.

Big Data analytics can identify customers who are likely to leave and help the company take preventive action.

Data Sources

The company can collect:

- Call records
- Internet usage

- SMS activity
- Recharge history
- Billing information
- Customer complaints
- Service requests
- Customer support interactions
- Plan details

Major Big Data Challenges

1. High Volume

Millions of customers can generate enormous quantities of interaction records.

Solution: Use distributed storage and scalable processing.

2. High Velocity

Call and network events are generated continuously.

Solution: Use real-time data ingestion and stream processing.

3. Data Variety

Customer information may exist in databases, logs, applications, and text-based complaint records.

Solution: Use data integration tools capable of handling multiple formats.

4. Data Quality

Duplicate, incomplete, or inconsistent customer information may affect prediction accuracy.

Solution: Apply data cleaning and validation.

5. Security and Privacy

Customer information is sensitive.

Solution: Apply authentication, authorization, encryption, and auditing.

Analytics Strategy

A churn analytics system can follow these steps:

Data Collection → Data Cleaning → Feature Creation → Churn Model → Customer Risk Score → Business

Action

Useful features may include:

- Frequency of complaints
- Number of service requests
- Recharge frequency
- Average monthly usage
- Billing delays
- Changes in data usage
- Customer tenure

Predictive Analytics

A predictive model can classify customers into categories such as:

- Low churn risk
- Medium churn risk
- High churn risk

Customers with high churn probability can receive targeted offers or improved support.

Integration Strategy

Data from billing systems, customer relationship systems, network logs, and support applications should be integrated into a common Big Data environment.

Batch processing can be used for historical analysis, while stream processing can be used for recent customer events.

Expected Benefits

The telecom company can:

- Identify high-risk customers
- Improve customer retention
- Reduce revenue loss
- Personalize offers
- Improve customer service
- Understand reasons for customer churn

Conclusion

Big Data analytics allows telecom companies to move from reacting to customer loss toward predicting possible churn. A combination of distributed storage, data integration, batch processing, real-time processing, and predictive analytics provides a scalable solution.

4. Smart City – Distributed File Systems and Data Integration

Question

A smart city project generates sensor streams from traffic, air quality, and surveillance systems. Explain how distributed file systems and data integration tools can support large-scale analysis.

Answer

A smart city generates data continuously from thousands of sensors and devices. Traffic sensors, air-quality monitors, surveillance cameras, parking systems, and other IoT devices produce large and diverse datasets. A distributed storage and integration architecture is required to manage this information.

Major Data Sources

The smart city may collect:

- Traffic sensor readings
- Vehicle information
- Air-quality measurements
- Surveillance video
- Weather information
- Parking data
- Public transportation information

Role of Distributed File Systems

A distributed file system divides large datasets into blocks and stores them across multiple machines.

Advantages

1. Scalability

More storage nodes can be added as sensor data increases.

2. Fault Tolerance

Copies of data can be maintained on multiple nodes so that failure of one node does not cause complete data loss.

3. Parallel Processing

Different parts of the dataset can be processed simultaneously.

4. Large-Scale Storage

Large sensor logs and surveillance datasets can be stored efficiently.

Role of Data Integration Tools

Data integration tools collect information from different systems and convert it into formats suitable for processing.

For example:

Traffic Sensors + Air Sensors + Surveillance + Weather → Integration Layer → Distributed Storage →

Analytics

The integration layer can perform:

- Data collection
- Data transformation
- Data cleaning
- Data filtering
- Data combination
- Data routing

Real-Time Analytics

Real-time processing can identify:

- Traffic congestion
- High pollution levels
- Accidents
- Unusual activities
- Parking availability

Batch Analytics

Historical data can be used to identify:

- Long-term traffic patterns
- Pollution trends
- Peak traffic hours
- Areas requiring infrastructure improvement

Proposed Architecture

Sensors → Data Integration/Ingestion → Distributed File System → Processing → Analytics → Smart City Dashboard

Example

If traffic sensors report increased vehicle density in a particular area, real-time analytics can identify congestion. The system can then provide information to traffic-management authorities so that traffic signals or routes can be adjusted.

Conclusion

Distributed file systems provide scalable and fault-tolerant storage, while integration tools collect and prepare data from multiple sources. Together, they enable large-scale smart-city analytics and support both real-time and historical decision making.

5. Financial Analytics Firm – Secure Big Data Solution

Question

A financial analytics firm must comply with strict privacy requirements while processing high-velocity transaction data. Design a secure Big Data solution and justify the controls chosen.

Answer

A financial analytics firm handles highly sensitive transaction information. Since transactions are generated continuously at high speed, the organization needs a system that provides both high-performance processing and strong security.

Proposed Architecture

Transaction Sources → Secure Data Ingestion → Encryption → Distributed Storage → Stream/Batch Processing → Analytics → Controlled Access

1. Secure Data Ingestion

Transaction data should enter the Big Data platform through authenticated and secure communication channels. Real-time ingestion is required because financial transactions are generated continuously.

2. Encryption

Encryption should be applied:

- **During transmission** – protects information while it moves through networks.
- **During storage** – protects information stored on disks or distributed storage.

Encryption reduces the risk of unauthorized disclosure.

3. Authentication

Users and applications should prove their identity before accessing the system.

Strong authentication mechanisms should be used for administrators and other privileged users.

4. Authorization

Authentication identifies the user, while authorization determines what the user is allowed to access.

For example:

- Analyst → Access to approved analytical datasets
- Manager → Access to business reports
- Administrator → System management privileges

Users should only receive the permissions necessary for their work.

5. Data Masking

Sensitive fields such as account numbers or personally identifiable information can be masked when full information is not required for analysis.

For example:

Original: 1234 5678 9012 3456

Masked: XXXX XXXX XXXX 3456

6. Audit Logging

The system should record important activities such as:

- User login
- Data access
- Data modification
- Administrative changes
- Failed access attempts

Audit logs help detect suspicious activity and support compliance investigations.

7. Data Classification

Financial data should be classified according to sensitivity. Highly confidential information should receive stronger controls than general business information.

8. Backup and Recovery

Regular backups and disaster-recovery mechanisms should be maintained. If a system failure or security incident occurs, critical data should be recoverable.

9. Real-Time Security Monitoring

Since transaction data arrives at high velocity, security monitoring should also operate continuously. Unusual transaction patterns can be identified using analytics.

For example, multiple unusual transactions from different locations within a short period may indicate suspicious activity.

Security Control Summary

Security Control	Purpose
Authentication	Verifies user identity
Authorization	Controls what users can access
Encryption	Protects data from unauthorized disclosure
Data Masking	Hides sensitive information
Audit Logging	Records access and system activities
Backup	Supports data recovery
Monitoring	Detects suspicious activities
Access Control	Limits data according to user roles

Conclusion

A secure financial Big Data solution should combine high-speed data ingestion and processing with strong privacy and security controls. Encryption, authentication, authorization, data masking, auditing, monitoring, backup, and controlled access provide multiple layers of protection. This approach allows the firm to process high-velocity transaction data while reducing privacy and operational risks.